

EXECUTIVE SUMMARY

Proactive **AI Engineer Intern** and Computer Science undergraduate at University of Transport Ho Chi Minh City (UTH). Hands-on experience developing and deploying practical AI/ML solutions, including **Retrieval-Augmented Generation (RAG)** with local LLMs, **Explainable AI (XAI)** for threat classification, and predictive **Ensemble regression models (XGBoost, Random Forest)** with Geo Intelligence features. Skilled in architecting containerized backend microservices using **Python (FastAPI)**, **Java (Spring Boot 3)**, **Docker**, and vector/relational databases.

TECHNICAL SKILLS

- AI & Machine Learning:** Python, PyTorch, Scikit-learn, XGBoost, Random Forest, RAG (Retrieval-Augmented Generation), Ollama (Llama 3.2), FAISS Vector Search, SentenceTransformers, DistilBERT, SHAP (Explainable AI), Optuna, Unsupervised Anomaly Detection (Isolation Forest, One-Class SVM, LOF).
- Backend & Architecture:** FastAPI, Java 17/21, Spring Boot 3, Spring Security 6, Spring Cloud Gateway, Kafka, Redis, WebRTC, WebSocket (STOMP/SockJS), RESTful APIs, Microservices Architecture.
- Databases & Storage:** MySQL, Microsoft SQL Server, Redis, FAISS (Vector Database).
- Frontend & Tools:** React, Next.js 14, JavaScript (ES6+), TypeScript, Tailwind CSS, Chrome Extension (Manifest V3), Docker, Docker Compose, Git, Linux/Bash, Jupyter Notebook, Postman.

FEATURED PROJECTS

DocuMind-AI – AI-Powered PDF Question Answering System (RAG) 2024

- Python, FastAPI, Ollama (Llama 3.2:3b), FAISS, SentenceTransformers, MySQL, Docker, Tailwind CSS
- Engineered an end-to-end **RAG** pipeline enabling conversational Q&A over PDF documents with local LLM streaming responses in Vietnamese and English.
 - Implemented semantic text chunking and high-dimensional semantic search using **SentenceTransformers** and **FAISS** (384-dim vector indexing).
 - Built asynchronous **FastAPI SSE streaming** endpoints and containerized the entire stack with **Docker Compose** for local offline inference.

MailGuard-AI – Email Security Assistant & Phishing Detector with XAI 2024

- Python, FastAPI, DistilBERT, Transformers, SHAP (XAI), Scikit-Learn, Chrome Extension (MV3), Docker
- Developed a real-time Gmail security extension (Manifest V3) classifying incoming emails into 4 categories (Normal, Notice, Spam, Phishing).
 - Fine-tuned **DistilBERT** and Scikit-learn models; integrated **SHAP (Explainable AI)** to explain risk triggers directly inside user email views.
 - Orchestrated multi-container Docker environments and handled client-to-backend API routing and model inference pipelines.

House Price Prediction AI – TP.HCM Real Estate Valuation 2024

- Python, XGBoost 2.0, Random Forest, Scikit-Learn, SHAP, Optuna, Flask, Pandas
- Built an Ensemble ML model (**XGBoost + Random Forest**) on ~40,000 TP.HCM property records with **Geo Intelligence** features (distance to CBD, Metro lines).
 - Designed an anti-leakage pipeline (**Smoothed Target Encoding, K-Fold OOF**), applied **Monotone Constraints**, and optimized via **Optuna**.
 - Deployed an interactive **Flask web app** featuring SHAP explainability and estimation confidence intervals.

Network Intrusion & Fraud Anomaly Detection (Unsupervised ML) 2024

- Python, Scikit-Learn, Isolation Forest, One-Class SVM, Local Outlier Factor (LOF), Pandas
- Researched and compared 3 unsupervised algorithms (**Isolation Forest, One-Class SVM, LOF**) for Network Intrusion Detection on **NSL-KDD** benchmark data.
 - Engineered automated threshold tuning (tune_threshold) to maximize F1-Score; achieved **93.39% Precision** and **0.9338 ROC-AUC** with Isolation Forest.

BICAP – Blockchain Clean Agriculture Traceability Platform 2024

- Java 21, Spring Boot 3.4, Spring Cloud, Kafka, Redis, Next.js 14, VeChainThor Blockchain, Docker
- Contributed **30+ commits** to an enterprise-grade agricultural supply-chain traceability platform powered by Microservices and **VeChainThor Blockchain**.
 - Refactored service architecture, designed centralized error handling for client-API communications, and developed the Farm Manager frontend with Next.js.

EDUCATION

University of Transport Ho Chi Minh City (UTH) 2022 – Present

Bachelor of Computer Science | Cumulative GPA: 3.0 / 4.0

Key Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, AI Model Development, AI Deployment, Data Science, AI Ethics, Database Systems.